FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

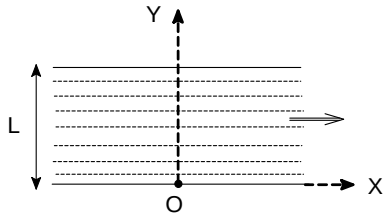
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A river is flowing with a variable flow rate given by $v = (v_0 + \alpha x)$. A swimmer starts from origin having speed v with respect to water wants to cross the river in minimum time. Then drift of swimmer along x – axis would be (width of river is L)



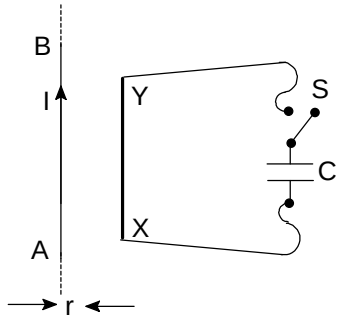
(A) $\frac{v_0}{\alpha}\left[e^{\frac{L\alpha}{v}} - 1\right]$

(B) $\frac{v_0}{\alpha}e^{\frac{L\alpha}{v}}$

(C) $\ln\left(\frac{v_0}{\alpha}\frac{L}{v}\right)$

(D) $\frac{v_0 L\alpha}{v}$

2. The figure shows a long fixed wire AB carrying constant current I. Another wire XY parallel to AB of mass per unit length λ is kept at a distance r from wire AB connected to a capacitor of capacitance C by flexible light wires through a switch. The capacitor is initially charged to voltage E_0. The switch is closed at t = 0 and the circuit has a negligible. The velocity of the wire XY, just after closing the switch will be



(A) $\frac{\mu_0 / CE_0}{4\pi r\lambda}$

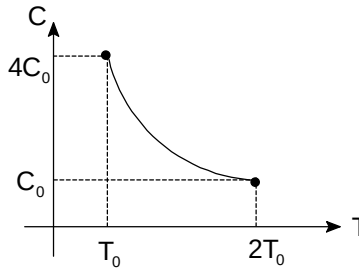
(B) $\frac{\mu_0 / CE_0}{2\pi r\lambda + L\mu_0 / C}$

(C) $\frac{\mu_0 / CE_0}{2\pi r\lambda}$

(D) $\frac{\mu_0 / CE_0}{4\pi r + L\mu_0 / C}$

Space for Rough work

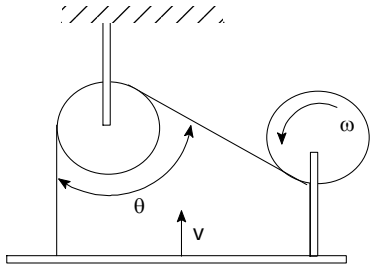
3. Find the heat energy required to raise the temperature of n motes of a gas from T_0 to $2T_0$ if its molar heat capacity varies as shown in the diagram. The equation of the curve is CT^x = constant



(A) $2nC_0T_0$ (B) nC_0T_0

(C) $\frac{nC_0T_0}{2}$ (D) $\frac{3}{2}nC_0T_0$

4. At the instant shown, the board is moving vertically upward with speed v. The drum of radius 'R' winds up at a rate $\omega(\text{rad/s})$. If the board always remains horizontal, springs taut. Find v.



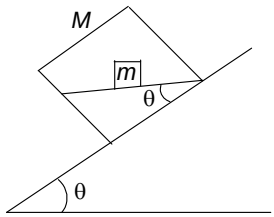
(A) ωR (B) $\frac{\omega R}{1+\cos\theta}$

(C) $\frac{\omega R}{\cos\theta}$ (D) $\frac{\omega R}{2+\cos\theta}$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. A box of mass M is kept on a frictionless and inclined plane which is inclined at an angle θ with the horizontal. A box is having another frictionless inclined plane, inclined at same angle θ with the previously described inclined plane as shown in the figure. If a block of mass m is kept on this inclined plane inside the box, then



(A) the acceleration of the block with respect to the box is $\frac{(M+m)g\sin\theta\cos\theta}{M+m\sin^2\theta}$ towards the left side.

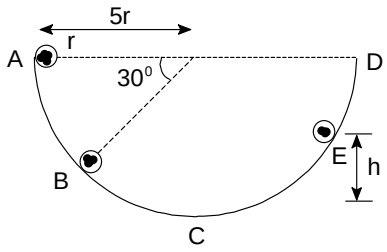
(B) the acceleration of the block with respect to the box is $\frac{(M+m)g\sin\theta\cos\theta}{M+m\sin^2\theta}$ towards the right side.

(C) the normal reaction between the box and the inclined plane is $\frac{M(M+m)g\cos\theta}{M+m\sin^2\theta}$.

(D) the normal reaction between the box and the block is $\frac{Mmg\cos^2\theta}{M+m\sin^2\theta}$

Space for Rough work

6. A solid sphere of radius r and mass m is released from point A on a vertical circular surface of radius 5r which has sufficient friction in part AB. Part BCD is smooth and h is the maximum height attained by the centre of the sphere from point C in part CD. The total energy and the kinetic energy at this height is T and K, respectively (assume that a lowest point C of the circular path as reference point for potential energy). Then



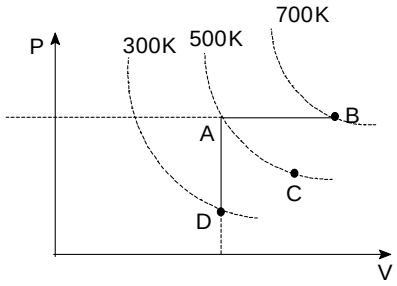
(A) $h = \left(\frac{31}{7}\right)r$.

(B) $T = 5mgr$

(C) $K = \left(\frac{4}{7}\right)mgr$

(D) The friction force on the sphere just before point B is $\frac{2\sqrt{3}mg}{5}$.

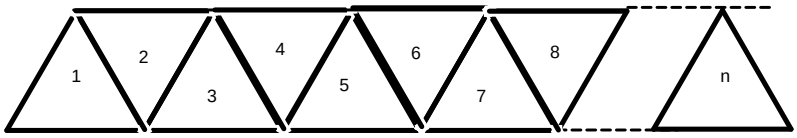
7. Figure shows three isotherms at temperatures 300K, 500K, and 700K, respectively, on PV diagram with broken lines. Three independent thermodynamic process (AB, AC and AD) are performed on 1 mole of monatomic ideal gas as shown in the figure. Then, choose the correct choice(s).



(A) In the process AC, first the temperature of the thermodynamic system increases and then decreases.
(B) In the process AC, the temperature of the thermodynamic system is always constant.
(C) In the process AD, average kinetic energy of molecules per unit volume decreases.
(D) None of these

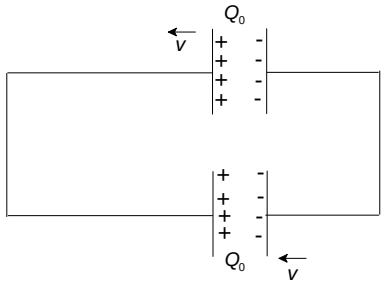
Space for Rough work

8. n number of identical equilateral prisms are kept in contact as shown in the figure. If the deviation through a single prism is δ, then (n, m are integers)



(A) if n = 2m, the deviation through the system of n prisms is zero.
(B) if n = 2m + 1, deviation through the system of n prisms is δ.
(C) if n = 2m, the deviation through the system of n prisms is δ.
(D) if n = 2m + 1 is deviation through the system of n prisms is zero.

9. Two identical capacitors are connected as shown and having initial charge Q_0. Separation between plates of capacitors is d_0. Suddenly the left plate of upper capacitor and right plate of lower capacitor start moving with speed v towards left while other plate of capacitor remains fixed.



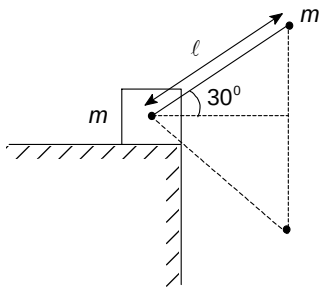
(A) The value of currents in the circuit be $\frac{Q_0 v}{d_0}$

(B) The value of current in the circuit be $\frac{Q_0 v}{2d_0}$

(C) The value of charge on lower capacitor at time t is $\frac{Q_0 v}{d_0} t$

(D) The value of charge on lower capacitor at time t is $\frac{Q_0 (d_0 + vt)}{d_0}$

10. A block of mass m(= 1 kg) is placed on a smooth horizontal surface as shown. Its centre is tied to a string of length ℓ (= 10 m) and other end of the string is connected to ball of same mass. Ball is released from the position shown. It falls under gravity and then string becomes taut. The kinetic energy of block and ball just after string is taut are KE_b and KE_p respectively, then



(A) $KE = 6\ J$ (B) $KE_p = 78\ J$

(C) $KE_b = 12$ J (D) $KE_p = 39$ J

Space for Rough work

11. An atom has energy levels $E_n = -\frac{A}{n^2}$ where n is an integer and A is a constant. Among the spectral lines that atom can absorb at room temperature are two adjacent lines with wavelengths 97.5 nm and 102.8 nm. Select the correct alternative(s)

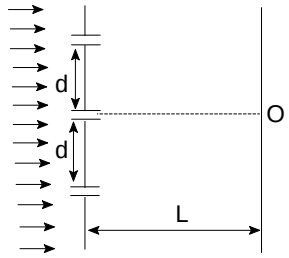
(A) $A = 54.4$ eV

(B) $A = 13.6$ eV

(C) The wavelength 97.5 nm corresponds to transition from n = 1 to n = 3

(D) The wavelength 102.8 nm corresponds to transition from n = 3 to n = 4

12. Suppose a monochromatic coherent, parallel beam of light, of wavelength λ passes three parallel slits each separated by a distance d from its neighbor. The distance of a point on the screen from the centre point O. at which a minima is formed $(L >> d)$



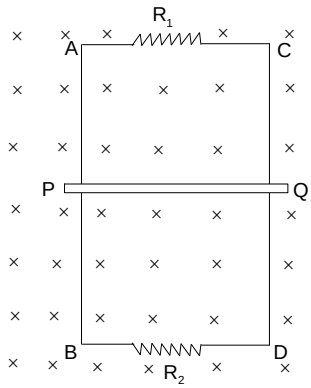
(A) $\frac{\lambda L}{3d}$ (B) $\frac{2\lambda L}{3d}$

(C) $\frac{4\lambda L}{3d}$ (D) $\frac{\lambda L}{d}$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

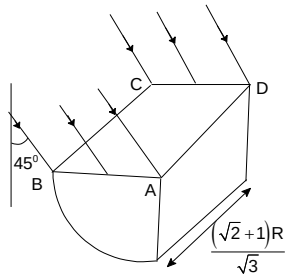
13. Two parallel vertical metallic rails AB and CD are separated by 1 m. They are connected at the two ends by resistances R_1 and R_2 as shown in the figure. A horizontal metallic bar PQ of mass 0.2 kg sides with out friction, vertically down the rails under the action of gravity. There is uniform horizontal magnetic field of 0.6 T, which is perpendicular to plane of the rails. It is observed that when the terminal velocity attained, the power dissipated in resistance R_1 and R_2 are 0.76 W and 1.2 W, respectively. Find the terminal velocity (in m/s) of the bar. (Take g = 9.8 m/s^2)



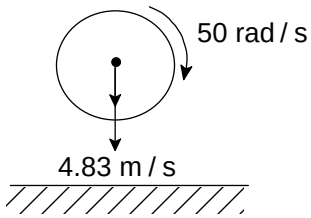
Space for Rough work

14. River stream velocity grows in proportion to the distance from the bank and reaches its maximum velocity 2 m/s in the middle. Near the bank, the velocity is zero. The speed of swimmer in still water is 5 m/s and directed perpendicular to the river stream. The width of the river is 100 m the drifting in swimmer is 5n metre. Find the value of n.

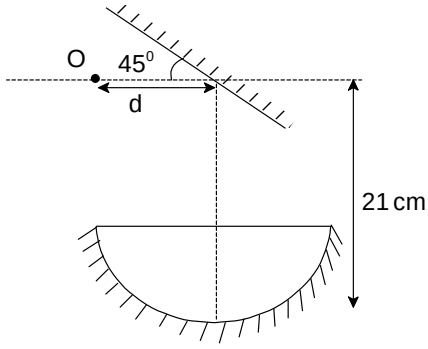
15. Light rays are falling on the flat surface ABCD of quarter of a cylinder of radius R = 3 m. Find the area (in m^2) on the flat surface on which falling light rays emit from the curved surface of the cylinder. Refractive index of the cylinder is $\sqrt{2}$.



16. A solid rubber ball of mass 2 kg and radius 3 cm collides with the horizontal ground with velocity 4.83 m/s and angular velocity 50 rad/s. The coefficient of friction and restitution are 2 and 0.5 respectively. Find the horizontal distance covered by the ball between the first and second impact of the ground in m.



17. A point object O is placed in front of a plane mirror and plano – convex lens (whose convex side is silvered) assembly, as shown. Find the distance d (in cm) of the object from the plane mirror so that the final image is created at the itself. Radius of curvature of the convex side of plano – convex lens is 42 cm and its refractive index is $\frac{3}{2}$.



18. Two candles A and B, each 30 cm long and requiring 4 and 6 hour respectively to burn out, stand vertically at a distance of 30 cm. The shadow of b falls on a vertical wall at a distance 3 m from b. Find the speed of the end of shadow (in decimeter/hour).

Space for Rough work

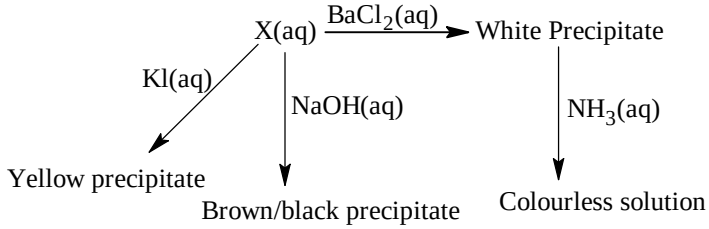
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. A compound 'X' reacts in the following ways

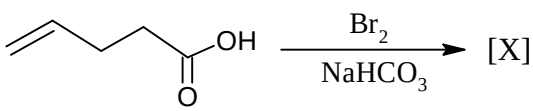


The compound 'X' is likely to be

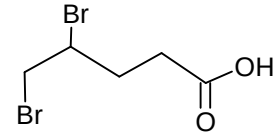
(A) $Pb(NO_3)_2$ (B) $CaCrO_4$

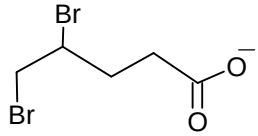
(C) $Hg(NO_3)_2$ (D) $AgNO_3$

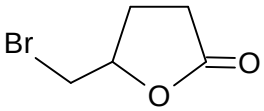
20.

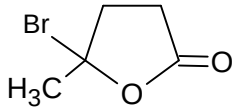


[X] is

(A) 

(B) 

(C) 

(D) 

21. $FeCl_3$ solution is treated with excess $NaOH$. A brown ppt. is formed. The ppt is filtered and dried finally heated with carbon monoxide at approximately 700^0C the products of the reaction is /are.

(A) $FeO + CO_2$ (B) $Fe_3O_4 + CO_2$

(C) only *Fe* (D) $Fe + CO_2$

Space for Rough work

22. The following reaction was carried out in water:

$Cl_2 + 2I^- \longrightarrow I_2 + 2Cl^-$

The initial concentration of I^- was 0.25 mol L^{-1} and the concentration after 10 minutes was 0.23 mol L^{-1}. Then the rate of appearance of I_2 is:

(A) $1\times10^{-3}\, mol\, L^{-1}\, \min^{-1}$ (B) $0.002\, mol\, L^{-1}\, \min^{-1}$

(C) $1\times10^{-2}\, mol\, L^{-1}\, \min^{-1}$ (D) $0.02\, mol\, L^{-1}\, \min^{-1}$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

23. Which of the following is true for 3-methyl-2-butanone?
(A) It may be prepared by CrO_3 oxidation of 2-methyl-2-butanol.
(B) Reaction with $NaBH_4$ gives a secondary alcohol.
(C) It may be prepared by acidic Hg^{2+} catalyzed hydration of 3-methyl-1-butyne.
(D) It forms a silver mirror on treatment with $Ag(NH_3)_2^+$.

24. Which of the following statements is/are correct?
(A) The order of splitting energy is $PtCl_4^{2-} > PdCl_4^{2-} > NiCl_4^{2-}$
(B) $[Co(NH_3)_6]^{3+}$ is colourless whereas $[Ni(H_2O)_6]^{2+}$ is coloured.
(C) $[M(en)(gly)]^{n+}$ will represent geometrical isomerism.
(D) The magnetic moment of $K_3[Fe(CN)_6]$ is $\sqrt{3}$ B.M.

25. Which of the following statement is/are incorrect:
(A) K_2CO_3 is more covalent than Na_2CO_3
(B) Cyanogen $(CN)_2$ is inert towards NaOH
(C) N_3^- and CN^- are pseudohalides.
(D) Br_2 does not disproportionate in alkaline solution

26. In Na_2O structure:
(A) O^{2-} ions constitute ccp.
(B) Na^+ ions occupy all the tetrahedral holes.
(C) Na^+ ions occupy all the octahedral holes.
(D) O^{2-} ions occupy half of octahedral holes.

Space for Rough work

27. In which of the following reactions an isocyanate is expected as an intermediate?

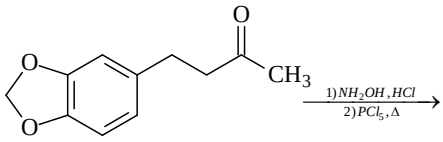
(A) $R-\overset{\overset{O}{\|}}{C}-NH\cdot NH_2 \xrightarrow[ii)\Delta,\ iii)\bar{O}H,H_2O]{i)HNO_2}$

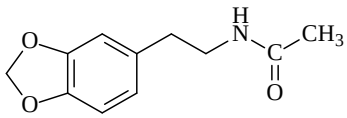
(B) $R-\overset{\overset{O}{\|}}{C}-OH \xrightarrow[iii)OH^-/H_2O]{i)HN_3,ii)\Delta}$

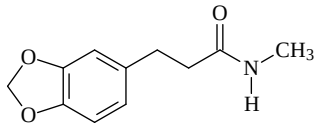
(C) $R-\overset{\overset{O}{\|}}{C}-NH\cdot OCO\cdot R \xrightarrow[ii)heat]{i)OH^-}$

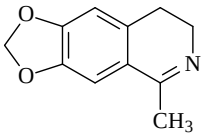
(D) $R-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{Br_2/\bar{O}H}$

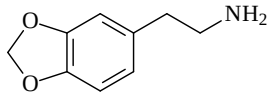
28. Which of the following will be obtained in the given reaction?



(A) 

(B) 

(C) 

(D) 

29. Kolbe's electrolytic method can be applied on

(A) CH_3COONa

(B) $\begin{array}{l}CH_2COOK\\|\\CH_2COOK\end{array}$

(C) $\begin{array}{l}HC-COOK\\\|\\HC-COOK\end{array}$

(D) C_6H_5COONa

30. For the given transition: C (graphite) $\longrightarrow$ C (diamond), correct options is/are

(A) C (diamond) is more thermodynamically stable then C (graphite) at 25°C

(B) C (graphite) is more thermodynamically stable than C (diamond) at 25°C

(C) diamond will provide more heat on complete combustion at 25°C

(D) $\Delta G_{transition}$ of C (diamond) $\longrightarrow$ C(graphite) is -ve

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

31. A polyhydric alcohol is acylated by using ethanoyl chloride in presence of alkali. It was observed that molecular mass of alcohol is increased to 237% on acylation. If the molecular mass of polyhydric alcohol is 92, then the number of –OH groups present in polyhydric alcohol are

32. 80 mL sample solution of KI was electrolyzed for 3.0 minute, using a constant current. The I_2 produced required 0.25 M, 37.2 mL sodium thiosulphate solution and unreacted KI required 36.3 mL 0.02 M acidic solution of $KMnO_4$. Determine the current (in ampere) passed during the electrolysis.

33. Two organic liquids A and B have vapour pressure p_1^0 & p_2^0 at 80^0C in their pure states. A mixture of the two boils at 80^0C and shows ideal behaviour. It has mole fraction of A = 0.16. If $p_2^0 - p_1^0$ = 472 mm; then p_1^0 will be (in mm of Hg):

34. Among the following, in how many number of colloids, particles of D.P. move towards cathode during electro-osmosis?
Ferric hydroxide, methylene blue , Au, arsenius sulphide, Haemoglobin, Starch, silicic acid , congo red , gelatin

35. How many milligram of CaC_2O_4 will dissolve in one litre of saturated solution?
(K_{sp} of CaC_2O_4 is 2.5×10^{-9} mol^{-2})

36. 0.535 g mixture of ethanol and acetaldehyde when heated with Fehling solution gave 1.2 g of a red precipitate. What is the percentage of acetaldehyde in the mixture?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If a, b, c are three positive real number such that $2a+b+3c=1$ then maximum value of $\left(ab^2c^2\right)$

(A) $\frac{8}{9(5)^5}$ (B) $\frac{8}{9(5)^4}$

(C) $\frac{8}{9(3)^5}$ (D) $\frac{8}{9(5)^3}$

38. A bag contain three biased coins, odds in favour of tail when tossed are $3:1, 1:1, 4:1$ respectively. A coin is drawn at random and tossed results into a tail, then the probability that if the same coin is tossed again will result into tail is

(A) $\frac{315}{469}$ (B) $\frac{193}{295}$

(C) $\frac{253}{600}$ (D) $\frac{381}{620}$

39. If $\int \frac{\tan x}{1+\tan x+\tan^2 x}dx = x-\frac{K}{\sqrt{A}}\tan^{-1}\left(\frac{K\tan x+1}{\sqrt{A}}\right)+C$ (c is constant of integration). Then the ordered pair (K, A) is equal to

(A) (2, 3) (B) (3, 2)

(C) (-2, 1) (D) (-2, 3)

Space for Rough work

40. The integral $\int_a^b \frac{e^{\frac{x^2}{a^2}} - e^{\frac{b^2}{x^2}}}{x} dx$ is equal to

(A) 0 (B) $\sqrt{\frac{b}{a}}$

(C) $\sqrt{\frac{a}{b}}$ (D) None of these

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. If the equation $x^4 - (a+2)x^3 + 2x^2 + (a+2)x - 3 = 0$ has at least two distinct positive roots then

(A) $a \in [1, \infty)$ (B) $a \in [2, \infty)$

(C) $a \in [2, 4]$ (D) $a \in \left[\sqrt{3} - 1, \infty\right)$

42. Two circles of radii r_1 *and* r_2 touch the x-axis at A and B respectively and the line $y = mx$ they intersect each other at $P(9, 6)$ *and* Q *also* $r_1 r_2 = 68$, then

(A) $m = \frac{12\sqrt{221}}{49}$ (B) $m = \frac{-12\sqrt{221}}{49}$

(C) $m = \frac{221\sqrt{2}}{49}$ (D) None

Space for Rough work

43. All value of $x \in \left(0, \frac{\pi}{2}\right)$ such that $\frac{\sqrt{3}-1}{\sin x}+\frac{\sqrt{3}+1}{\cos x}=4\sqrt{2}$ is/are

(A) $\frac{\pi}{15}$ (B) $\frac{\pi}{12}$

(C) $\frac{3\pi}{10}$ (D) $\frac{11\pi}{36}$

44. If $\vec{a}$ *and* $\vec{b}$ are two vectors such that $\left|\vec{a}+\vec{b}\right|=5\sqrt{2}$ *and* $\vec{a}\times\left(4\hat{i}+\lambda\,\hat{j}+2\hat{k}\right)=\left(4\hat{i}+\lambda\,\hat{j}+2\hat{k}\right)\times\vec{b}$ and $\left(\vec{a}+\vec{b}\right)\cdot\left(-3\hat{i}+2\,\hat{j}+\lambda\hat{k}\right)=10$ then

(A) Sum of values of $\lambda=\frac{48}{7}$ (B) Product of value of $\lambda=\frac{84}{7}$

(C) No possible value of λ (D) Two possible value of λ

45. Let $f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2}$ for all real x and y. If $f'(o)$ exists and equal to -1 and $f(o)=1$ then

(A) $f(x)=1-x$ (B) $f(x)=1-x^2$

(C) $f(2)=-1$ (D) $f(2)=1$

46. The locus of a point whose chord of contact with respect to the ellipse $x^2+2y^2=1$ subtend a right angle at the centre of the ellipse is

(A) $x^2+4y^2=3$ (B) $y^2=4x$

(C) $2x^2+4y^2=3$ (D) $x^2=4y$

Space for Rough work

47. If $A=\begin{bmatrix}1 & 0 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}$ then

(A) $A^3-A^2=A-I$

(B) $\det\left(A^{2010}-I\right)=0$

(C) $A^{50}=\begin{bmatrix}1 & 0 & 0\\ 25 & 1 & 0\\ 25 & 0 & 1\end{bmatrix}$

(D) $A^{50}=\begin{bmatrix}1 & 1 & 0\\ 25 & 1 & 0\\ 25 & 0 & 1\end{bmatrix}$

48. If α, β, γ are roots of the equation $9x^3-7x+6=0$ then the equation $x^3+Ax^2+Bx+C=0$ has the roots $3\alpha+2, 3\beta+2, 3\gamma+2$ then

(A) $A=6$

(B) $B=-5$

(C) $C=24$

(D) $A+B+C=23$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

49. Let $L_1:\frac{x-6}{3}=\frac{y-7}{-1}=\frac{z-4}{1}$ *and* $L_2:\frac{x}{-3}=\frac{y+9}{2}=\frac{z-2}{4}$. *If* L_1 *and* L_2 are skew lines such that P and Q are two points nearest to each other lying on lines L_1 *and* L_2 respectively. If image of P with respect to plane $3x+3y-z=11$ *is* $R(a, b, c)$ *then* $-a+b+\frac{c}{2}$ is

50. If x and y are the solution of the equation $12\sin x+5\cos x=2y^2-8y+21$ then the value of $6\cot\left(\frac{xy}{2}\right)$ is

Space for Rough work

51. If $p(x) = x^4 + ax^3 + bx^2 + cx + d$ (where a, b, c, d∈R) and $p(1) = 10,\ p(2) = 20,\ p(3) = 30$

then the value of $\left(\frac{p(12) + p(-8) - 10}{10000}\right)$ equals to

52. If $\frac{\tan 8^\circ}{1-3\tan^2 8} + \frac{3\tan 24^\circ}{1-3\tan^2 24} + \frac{9\tan 72^\circ}{1-3\tan^2 72} + \frac{27\tan 216^\circ}{1-3\tan^2 216} = x\tan 108^\circ + y\tan 8^\circ$
then $\left(x^2 + y^2\right)$ is equal to

53. If $A = \lim_{n\to\infty}\left({}^nC_0 \cdot {}^nC_1 \cdot {}^nC_2 \cdot {}^nC_3 \cdots {}^nC_n\right)^{\frac{1}{n(n+1)}}$ then A^2 is equal to

54. The Area bounded by the curve $y = \sin^{-1}|\sin x| + \cos^{-1}(\cos x)$ *and* $y = \cos x + \pi$ and the lines $x = 0 \quad x = 4\pi$ is

Space for Rough work